

Assignment #1.

Question #2:-

Target Machine: "Self Checkout Machine at a Grocery Store"

Target Users: Everyday shoppers across diverse age groups, abilities & tech-literacy levels.

Core Goal: Scan, bag and pay for purchased items quickly without waiting in a cashier line.

User Experience:-

1. Goal: Buy items quickly without waiting for the cashier.
2. Plan: Scan items one by one, place them in the bagging area and pay by card.
3. Action: Swipe an item's barcode across the glass scanner.
4. Machine Response: Beeps, flashes a light and weighs the item.
5. Information delayed: Shows item name, price, running total, & next steps.
6. Understanding: Clear for regular items; confusing for produce or weight errors.

7. Success / Failure: Success shows a thank you screen & prints a receipt.
Failure flashes an error light & locks the screen.

Frustrating Points:-

- Sensitive weight scales lock the screen over minor bag movement.
- Unbarcoded produce requires deep searching through confusing menus.
- Error screens do not tell the problem without an attendant.

Physical / Ergonomic Problems:

- Fixed screen height:- Hard to reach for wheel chair users.
- Small payment keypad:- hard to see and press for visually impaired users.
- Cramped layout:- small shelves make scanning & bagging large orders physically awkward.

Improvements:-

- Adjustable screens:-
 - Add a tilting screen mount and anti-glare screen coating.

Benefits → Reduces light reflection and lets users of any height see clearly.
limitations → Vision & biomechanics.

• AI Camera & Visual Error Prompts:-
→ Add an overhead camera to auto-identify, produce and display clear fix-it instructions on screen.

Benefit → Removes code searching and lets users fix simple errors themselves.
limitations → Memory & cognitive load.

Question # 3:-

System Name: NavSight (AR smart Glasses for the visually impaired).

Domain: Assessability and Navigation.

System Overview & Core Interaction:

- Problem Solved: Navigational independence.
- Target Users: People who are blind, low visioned, vision loss.
- Input methods: Computer Vision Camera, LiDAR depth Sensors, special microphone array for voice commands etc.

→ Output methods: Directional 3D audio, localized tactile vibration motors built into the frame arms, and low-latency audio descriptions.

→ Primary Interaction Style: Pervasive AI paired with Agent based Interface.

AHCI interaction Style :-

The system operates quietly in the bg without requiring continuous manual interaction.

It is better than existing solutions because it combines micro-hazard detection, real-time spatial mapping & natural conversational AI.

Ethics, limitations and Accessibility :-

→ Privacy & Ethical Risks :-

1. Bystander Privacy:

Continuous camera capture in public spaces can collect faces and personal details without consent.

2. Safety Dependency:

Algorithmic failure or false negatives could lead to physical injury

→ Technological Limitations:-
Real time processing & Power/heat constraints.

→ Accessibility Features:-

- Hands free control
- Customizable Audio Cues.
- Haptic Directionals.

Panel 1

[User facing Problem]

User while came approaching an unseen low hanging branch.



Panel 2

[User starts System]

User taps new sight frame and speaks:-
"Navigate to metro"



Panel 3

[System responding]

Glasses scan tree branch ahead.
Spatial audio: "low branch 3 feet ahead"



Panel 4 [User Receives Result]

User ducks and steps left to bypass the branch.



Panel 5 [Successful outcome]

User smoothly walks past the hazard.

Question # 1.

Selected App: Youtube

1. Interaction Styles:

- Direct manipulation: swiping down on a video opens a mini player.
- Menu-Driven interaction: Users select predefined options from structured list rather than executing complex commands.
- Gestures: Double-tapping the right or left side of the video fast-forwards.

2. Task Evaluation: Sharing a video w/ a friend

- Clarity: exceptionally easy with prominent "share" btn.
- Interaction flow: The single tap and clear labeling minimize cognitive effort, making execution virtually effortless.
- System feedback: Very easy to evaluate with several "toast notifications" that appear after every mini task.

3. Research Paper Analysis

Title: "Design Strategies to Minimize mobile Usability Issues in Navigation."

Authors: H.S.A. Al-Nafjan et al. (2024)
Citation: Information, 15(11), 732.
<https://doi.org/10.3390/info15110732>.

Main findings :-

The study evaluates mobile navigation usability issues across popular media applications, including Youtube, through empirical testing and System Usability Scale (SUS) measurements and researchers found that existing, overly hidden or non-standard bottom-nav patterns cause user frustrations & errors.

Relation to HCI principles:

- Recognition over Recall: showing explicit text labels alongside icons removes the need to memorize abstract icons.
- Visibility of System Status: clear, persistent bottom navigation anchors the user within the System architecture.